

Ashfakur Rahman

Adjunct Faculty, Dept. of CSE, Leading University, Sylhet
MSc Student in Data Science, Dept. of Statistics SUST, Sylhet
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Research & Teaching Profile

I am a dedicated Computer Science and Engineering graduate from Sylhet Engineering College (CGPA: 3.89/4.00) with a genuine passion for teaching, research, and academic contribution. Currently serving as an Adjunct Faculty at Leading University and concurrently pursuing an MSc in Data Science at Shahjalal University of Science and Technology (SUST), I bring both theoretical depth and practical classroom experience to the role of a Lecturer. My teaching philosophy centres on building conceptual clarity and encouraging independent problem-solving — values I have actively applied while instructing undergraduate students in theory and laboratory courses.

My research interests lie in Machine Learning, Natural Language Processing, and Geospatial Deep Learning, with a particular focus on low-resource Bengali NLP and intelligent document understanding. I have contributed to two IEEE-published papers and am currently working on an MSc thesis on urban plastic waste detection using multi-modal deep learning. I am committed to combining rigorous research with impactful pedagogy, and I am eager to contribute meaningfully to an academic department through quality teaching, scholarly publications, and student mentorship.

Education

- Shahjalal University of Science and Technology (SUST), Sylhet** 2025 – Present
M.Sc. in Data Science (Professional) Currently Enrolled
- Sylhet Engineering College (SEC), Sylhet** 2019 – 2024
B.Sc.(Engg.) in Computer Science & Engineering CGPA: **3.89** / 4.00
- Kanaighat Govt. College, Kanaighat, Sylhet** 2017 – 2018
Higher Secondary School Certificate (HSC) GPA: 4.92 / 5.00
- Kanaighat Govt. High School, Kanaighat, Sylhet** 2015 – 2016
Secondary School Certificate (SSC) GPA: 5.00 / 5.00

MSc Thesis (Ongoing)

- Title:** *Geospatial Deep Learning for Urban Plastic Waste Hotspot Detection and Accumulation Forecasting: A Multi-Modal Approach for Bangladesh*
- Supervisor:** Dr. Md Forhad Rabbi, SMIEEE — Professor & Head, Dept. of CSE, SUST
- Developing a multi-modal deep learning framework integrating geospatial data, satellite imagery, and environmental factors.
- Focused on hotspot detection, spatial analysis, and predictive modeling for sustainable urban waste management.

Publications

Published

- Rahman, H., Rahman, A., Islam, A., Raki, S.A., & Mojumder, M.A.N. (2024). **BDeedNet: A Deep Learning Framework for Bengali Deed Summarization**. In *Proceedings of ICCIT 2024*. IEEE.
- Mojumder, M.A., Islam, T., Rahman, A., Das, S., & Chowdhury et al. (2023). **Bangla and Nagri Language Detection and Classification Using Deep Learning Methods on Text Document Image**. In *Proceedings of ICAECIS 2023*. IEEE.

In Preparation (manuscripts in preparation for journal/conference submission)

- Rahman, A. et al. *Large Language Models (LLMs) for Low-Resource Bengali NLP: Fine-tuning and Prompt Engineering for Improved Performance in Real-World Applications*.
- Rahman, A. et al. *Retrieval-Augmented Generation (RAG) for Bengali Question Answering: Enhancing Factual Accuracy and Reducing Hallucination in Domain-Specific Tasks*.
- Rahman, A. et al. *Multimodal Document Understanding: Combining Text and Visual Features for Intelligent Analysis of Bengali Legal and Structured Documents*.

Teaching Experience

• Adjunct Faculty

Fall 2024 – Present

Leading University, Department of Computer Science & Engineering

Sylhet, Bangladesh

- Delivered theory and laboratory instruction across **5 consecutive semesters** (Fall 2024 – Summer 2026), teaching multiple sections per semester across a broad range of undergraduate CSE courses.
- Courses taught include: **Information Security & Cryptography, Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems, Data Communication, Computer Graphics, Microprocessor, Management Information Systems, Introduction to Computing, and Computer Programming-I.**
- Designed course materials, assignments, and assessments aligned with curriculum objectives; employed problem-based teaching strategies to strengthen students' analytical and programming skills.
- Provided continuous academic mentorship, evaluated student performance across multiple sections, and maintained a collaborative and intellectually stimulating classroom environment.

• Part-Time Teacher

July 2023 – December 2023

Israb Ali High School & College

Dakshin Surma, Sylhet

- Taught **Information and Communication Technology (ICT)** and **Higher Mathematics I & II** to HSC (Class 11–12) students across theory and practical sessions.
- Planned and delivered structured lessons with a focus on conceptual clarity; prepared assignments and model questions aligned with the national curriculum.
- Conducted class tests and terminal examinations, evaluated answer scripts, and provided detailed feedback to support academic improvement.
- Offered individual academic guidance to students, helping them build confidence and develop strong foundational skills in mathematics and computing.

Academic Projects

• Pharmacy Management System

[Source Code](#)

Under the Object-Oriented Programming Course

- **Tools:** Java, Apache NetBeans / Eclipse, XML
- Streamlines pharmacy operations and doctor appointments. Manages drug inventory, ensures sufficient stock, detects drug interactions, and facilitates scheduling.
- Supervised by Nayan Kumar Nath, Lecturer, Dept. of CSE, SEC.

• Department Management System

[Source Code](#)

Under the Software Development Course

- **Tools:** Python, Django, jQuery, SQLite, HTML, CSS, Bootstrap
- Manages departmental activities with an HTML/CSS/Bootstrap front-end and a Python/Django/SQLite back-end.
- Supervised by Nayan Kumar Nath, Lecturer, Dept. of CSE, SEC.

• Fake News Detection

[Source Code](#)

Under the Machine Learning Course

- **Tools:** Python, Scikit-learn, SVM, Naive Bayes, HTML, CSS, Jupyter Notebook
- Machine learning-based fake news detection application utilizing SVM and Naive Bayes classifiers for robust news classification.
- Supervised by Md. Abu Naser Mojumder, Assistant Professor & Head, Dept. of CSE, SEC.

Technical Skills

Programming	Python, C, C++, Java
Data Science / ML	Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy
NLP / Deep Learning	Hugging Face Transformers, mT5, bT5, BERT
Visualization	Streamlit, Matplotlib, Seaborn
Web / Frameworks	Django, HTML, CSS, Bootstrap
Developer Tools	Git, GitHub, Jupyter Notebook, VS Code
Databases	MySQL, SQLite

Activities & Achievements

- **Runner-Up**, Intra-University Hackathon, Sylhet Engineering College
- **Founding Member**, SEC Research Lab
- **Organizing Secretary**, CSE Research Club, Sylhet Engineering College

- **Treasurer**, SEC Programming Club

Until August 2024

- **Organizing Committee Member**, SEC Inter-University Junior Programming Contest (IUJPC) 2022

- **Member**, SEC Developers Club

- **Member**, SEC-CSE Society

Until August 2024

References

Dr. Md Forhad Rabbi, SMIEEE

Professor & Head of CSE Department

Shahjalal University of Science and Technology, Sylhet

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Md. Abu Naser Mojumder

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